Fixed genus, unbounded fields: arithmetic counterfamily

Proved family. For every t≥0, the actual field F of size 3^(2t+1) has monic, separable degree-five polynomial X^5-X, with reduced-pair degree bound2. Field sizes are unbounded. The actual coefficient image of the complete F3 Cantor group has8elements; its concrete cyclic subgroup generated by (X,0) is proper and has order2. Both subgroup pairs are smooth, with fraction1. All ambient reduced pairs form a finite universe with smooth fraction strictly below1. Signed standing unverified. [pcc:unverified]

Same quintic X^5-X, reduced degree bound2; q=3^(2t+1) → ∞

Actual injective coefficient map; every image sum has a unique Cantor output.

Actual cyclic subgroup: order2

Generator (X,0); smooth fraction1

All ambient reduced polynomial pairs

Witness (X^2+1,0); smooth fraction<1

Control: q=3^(2t+2), same witness smooth. Full ambient Picard/group bridge open.

Actual arithmetic. Coefficient extension is proved injective and preserves monicity, degrees, separability and the reduced-pair equations. Actual normalized gcd naturality is derived from monic Bézout/divisibility witnesses, rather than confusing it with an unnormalized Euclidean gcd. Full gcd, composition numerator, exact quotient and reduction certificates are transported. Every group sum is equivalent to the unique actual normalized Cantor output, even against any ambient candidate. [pcc:unverified]

Full ambient counting domain. The ambient set contains every monic u of degree≤2 and v of degree below u with u dividing $v^2-(X^5-X)$. A bounded coefficient encoding proves finiteness; the set is not an eight-pair image list. The actual image witness $u=X^2+1$,v=0 does not split because card F≡3mod4 makes −1 nonsquare. A strict subtype cardinality inequality gives the ambient fraction<1. Lean proves unequal fractions beyond every field-size bound and directly negates universal equality on this family. [pcc:unverified]

Boundary control. Over every actual even-degree field of size 3^(2t+2), −1 is square. Lean explicitly factors X^2+1 and proves the same point7 smooth. A direct universal negation disproves preservation of its nonsmoothness under every finite coefficient extension. The odd-degree restriction has an actual positive/negative control. [pcc:unverified]

Scope. The group law is on the coefficient image, with actual Cantor compatibility. A full group law on all ambient pairs and the geometric Picard/Jacobian identification remain open. The subgroup order stays2. Strict finite disparities for unbounded fields do not establish a uniform limiting gap. They do not refute Gaudry's almost-prime-order/large-prime-subgroup hypothesis or the fixed-genus whole-Jacobian 1/g! asymptotic. [pcc:unverified]

Checks. Whole gate 60 declarations / 0 failures / 0 trusted native exceptions; Lean4.35.0, Mathlib56a00585…. Complete closed declaration, custom imports/configs and artifact pins are in sources.json. Failed/successful source/log/exit records and full gate are losslessly archived. Only propext/Classical.choice/Quot.sound; no holes, custom axioms, native evaluator, raised limits or new global elaboration flags. Exact PCC replay is separate; signed standing unverified. [pcc:unverified]

Sources. Gaudry2000 Definitions3/4, Section2.3 and Proposition4/proof; Stoll2014 Lemmas4.16–4.17/Theorem4.18 and Lemma5.2/proof, printedp30 visually inspected. Branch-factor torsion is established prior work; no novelty or qualified-source refutation is claimed. KB unavailable; bounded primary/repository/library fallback before edits. Other all-fourteen obligations remain in the current map. No new native experiment, sampling, paid compute or agent upload. [pcc:unverified]